anonymity of a dynamic link of a dial-up connection or the firewall of a network, you are pretty safe.

This may be true to some extent for intruders aren't exactly searching for your particular machine. But when the reviewers at *Network Computing* (India) conducted a surprise scan through some VSNL dial-up IP pools in Mumbai, they found that 89 out of around 2,000 IP addresses were ailing from Back Orifice infection. In addition, around 24 NetBus infections were identified. Another test they conducted involved nuking and pinging (*see Glossary*) the very same pool of IP addresses using two different DoS (Denial of Service) tools. They succeeded in around 50 per cent of the cases.

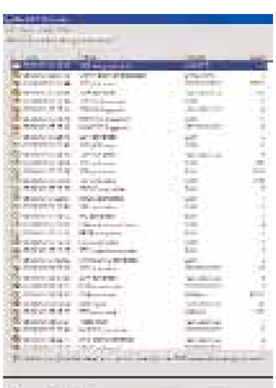
Now if your's was one of the machines under attack, you must have wondered what you did wrong to suddenly find your machine crashing. So you see, it is not necessary for a person to actually know you, or your IP address, to gain access to your machine. If your machine is infected or prone to exploits, then even beyond the anonymity of a dynamic IP address, your PC can be hacked. A personal firewall, therefore, is one of the methods you can use to deny access to such intrusions. So, let's get personal with the firewalls.

TEST PROCESS

The Test Bench in which the firewalls were installed consisted of a Pentium 266 MHz processor with 64 MB RAM. The operating system in test was Windows 98SE. TCP/IP and NetBEUI protocols were installed on the Test Bench and File and Print Sharing was also enabled before installation of the firewalls. The security level in IE 5.5 was medium with prompt before using any Java applets or ActiveX controls.

To gauge the application-level filtering features of the firewalls, we installed certain standard software that try to access the Net. Using these software, we tried to send files as well as download files from a testing machine. Apart from these software, a popular Trojan, SubSeven 2.2 server, was also installed on the Test Bench.

Testing tools: Using a test machine, we ran certain port scanners on the Test Bench. The *nix-based port scanner, Nmap (version 2.53), and Windows NT/2000 based Retina Scanner (version 3.02 beta) were used to scan the Test Bench. With both the scanners, the machine was first scanned before installing any firewall and then scanned after the firewall was installed.



Know who is intruding into your computer with BlackICE Defender

Apart from these port scanners, we tried accessing SubSeven Server on the Test Machine using the SubSeven client. At the same time, we also used certain application exploits such as those for ICQ. To test the ability of the firewall to detect ActiveX exploits, we went to certain sites that, using ActiveX controls or Java applets, uploaded certain files or read certain files from the local machine.

We also created a test HTML page with an ActiveX control embedded that installed the Thing Trojan on to the machine. We then tried opening the HTML page using IE and noted the behaviour of the firewall. Apart from these specific tools, we also tried nuking the Test Machine (sending continuous packets of data). The ability of the firewall to detect and ward off such intrusion attempts was noted.

In all, we tested 10 personal firewalls for their features and performance. And since getting the best out of your personal firewall requires patience and learning, ease of use or configuration settings were accorded the same importance as the actual features and performance. The features and performance as measured against the price gives the value for money for the firewall. The best firewall is therefore the one that has the maximum features and gives the best performance at a price that is easily affordable.

Star performers

What makes for a good performing firewall? The answers are simple—the ability to ward off all intrusion attempts, control applications that can

Porting Problems: How Firewalls Work

Firewalls basically work by acting as a filter between your Internet connection and your application. At a very basic level, you can grant or block rights for different applications to gain access to the Internet. But this is just one part of the working and is related to your access to the Net. Firewalls also work in a reverse manner by blocking access to outside traffic, thus managing to ward off attacks. This is done by monitoring the ports and the protocols that the outside application is trying to use. The main protocols used for Internet access, TCP and UDP, use port numbers to distinguish between different logical channels on the same network interface card. There are 65,536 logical ports possible and these port numbers can be seen as a door to the outside world. Some doors are closed, some are open, so access to certain services or protocols (HTTP, FTP) depends on the open/closed nature of ports. Firewalls again act as gatekeepers and, as per your rule-set, show a port as open or closed for communication. Hence, you can block ports that you don't use or even block common ports used by Trojans.

Using firewalls, you can also block protocols, so restricting access to NetBIOS will prevent computers on your network from accessing your shared folders. Firewalls often use a combination of ports, protocols, and application-level security to give the best results.

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More Info on
the test process and the products tested
Software
NeoWatch, Sygate and ZoneAlarm
Mindware

Contents